

Sheng Lu (Lucas Lu)

B.Eng. Candidate in Industrial Engineering | Class of 2029

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RESEARCH INTERESTS & MENTORSHIP

- Interests:** Operations Research & Optimization; Emergency Response & Logistics; Machine Learning & Data Science; Mathematical Modeling.
- Academic Mentor:** Professor Yu Wang, Department of Industrial Engineering, SUSTech.

EDUCATION

Southern University of Science and Technology (SUSTech)

B.Eng. in Industrial Engineering, Zhicheng College | Shenzhen, China

Aug. 2025 - Present

- GPA:** 3.91 / 4.0 | **Ranking:** 1 / 11.
- Selected Coursework:** Mathematical Analysis I-II (93, 96), Ordinary Differential Equations B (98), Foundation of Probability Theory (94), Linear Algebra (89), Introduction to C Programming (96), College Physics II (90).

SELECTED RESEARCH & PROJECTS

EMS Station Location Optimization

Project Lead | 120 EMS Station Location Optimization

2026 - Present

- Leading a 6-member team to optimize emergency medical service station placement across Shenzhen. Partitioned the city into 5,279 500m x 500m grids and developed an XGBoost ETA model using 620K+ dispatch records and 360K Tencent Maps samples (MAE = 1.64 min, R2 = 0.863); achieved 90% coverage within 10 minutes.
- Finalist, 8th China University Mechanical Engineering Innovation & Creativity Competition (2026). Tools: Python, OSMnx, XGBoost.

AED Deployment at Gulongzhong

Project Participant | AED Deployment at Gulongzhong

2026 - Present

- Comparing static AED placement with a dynamic human-vehicle-drone coordinated scheme using road-network analysis; exploring sustainable operation through insurance partnership models.

Meal Planning and Nutritional Optimization

Project Experience | Meal Optimization

2026

- Formulated an optimization framework for meal planning and food allocation under nutritional, budget, and variety constraints. Explored linear and integer programming to balance daily nutrient targets with practical menu choices and analyze cost-nutrition-diversity trade-offs.

SELECTED HONORS & AWARDS

- APRU ULP Outstanding Student Ambassador** (one of ten university-wide), APRU / SUSTech - 2026
- Second Prize**, 17th National College Student Mathematics Competition (Non-Math A) - 2025
- Outstanding Individual**, SUSTech Winter Social Practice - 2026
- Outstanding Camper**, 4th Xiancheng Program, Zhicheng College - 2025
- Social Impact Award**, Hundreds, Thousands, Myriads Project, Chengguang Volunteer Team - 2025
- Outstanding Volunteer Service Organization of 2025**, Chengguang Volunteer Team - 2025

LEADERSHIP & SERVICE

Core Student Leader | SUSTech IE Hunt

Optimization challenge planning and student team coordination

Oct. 2025 - Present

- Co-led Seasons 1-2 of a campus-wide optimization challenge, integrating the Traveling Salesman Problem and Linear Programming into game mechanics; coordinated a 5-member team across logistics, feasibility analysis, and promotion.

Student Representative | Zhicheng College 10th Anniversary Ceremony

Selected as the sole freshman representative to deliver a formal keynote speech

Oct. 2025

Group Leader | Orange Light Volunteer Service Team

Community research and accessibility advocacy

2025 - Present

- Led community research across Shenzhen, Foshan, and Guangzhou; partnered with the Shenzhen Association for the Blind on accessibility supervision and hosted Oxford University delegations at SUSTech.

SKILLS

- Programming & Technical:** Python, MATLAB, R, C, LaTeX, AnyLogic, XGBoost, scikit-learn, NumPy, Pandas, Matplotlib, OSMnx.
- Languages:** Mandarin (Native), English (Proficient), Korean (Daily basics).